24ADC001 — BUSINESS INTELLIGENCE
LAB ACTIVITY — TEAM MINI-PROJECT (AI-ASSISTED)

ChargeGrid: An AI-Assisted BI Mini-Project with OpenAI Tools

Duration: 2 hours | Team size: 2 | Concepts covered: BI/DSS pipeline, decision types (structured / semi-structured / unstructured), OLAP-style roll-up & drill-down via natural language, forecasting basics, tool-augmented knowledge work.

Tools required: ChatGPT with Advanced Data Analysis (Code Interpreter) for the analysis phase, and ChatGPT's image tool (DALL·E) for the dashboard mockup. Free-tier ChatGPT with a temporary Plus/edu seat, or any equivalent OpenAI-tool access already provisioned for the class, works.

Scenario

ChargeGrid operates EV charging stations across Coimbatore, Chennai, and Bengaluru. The operations head has one question for the next budget cycle: "where should we add fast chargers next quarter, and why?" Nobody has actually analyzed the session logs — they just glance at daily totals. Your team has two hours to turn the raw session data into a decision-ready recommendation, using AI tools to do the heavy lifting a human analyst would normally spend a day on.

Dataset: chargegrid_sessions.csv

1,800 charging sessions across the three cities over a 90-day window, with real embedded patterns for teams to uncover: subscriber commuters cluster around morning/evening slots on weekdays; pay-as-you-go users run longer sessions on weekends; and one city shows a distinct evening congestion signature that the data itself should reveal.

- Columns: session_id, date, city, station_id, charger_type (AC Slow / DC Fast), connector_type, user_type (Subscriber / Pay-as-you-go), energy_kwh, duration_min, cost, day_of_week, time_slot, weather.

Activity Structure (2 hours)

Time	Phase	What teams do
0:00 – 0:10	Kickoff	Form pairs, read the scenario, download chargegrid_sessions.csv, open ChatGPT with Advanced Data Analysis enabled.
0:10 – 0:55	Phase 1 — AI-Assisted OLAP Analysis	Upload the CSV to ChatGPT (Code Interpreter). Prompt it to roll up total sessions and average duration by city and month, then drill down to weekly and time-slot level for one city. Ask it to slice by user_type and dice by city + time_slot + weather. Task: find the one anomaly in the data that a manager would actually act on, and have ChatGPT generate the supporting chart.
0:55 – 1:30	Phase 2 — Decision Framing	Switch to plain ChatGPT chat. Prompt it to classify "where to add fast chargers next quarter" as structured, semi-structured, or unstructured, with justification. Ask it to draft three candidate recommendations with a pros/cons table, grounded only in the anomaly your team

		found in Phase 1 — reject any recommendation the data doesn't actually support.
1:30 – 1:50	Phase 3 — Dashboard Mockup	Use ChatGPT's image tool to generate one executive-dashboard mockup image showing the 3–4 KPIs your team would put in front of the ops head. Write a one-line caption for each KPI explaining why it belongs on the dashboard.
1:50 – 2:00	Wrap-up	Compile the one-page executive brief (see Deliverables) and submit. Two teams share their anomaly + recommendation with the class in one sentence each.

Deliverables (one submission per team)

- A one-page executive brief containing: the anomaly found in Phase 1 with its supporting chart; the decision classification and reasoning from Phase 2; the recommended action with trade-offs; the dashboard mockup image with KPI captions.
- An appendix listing the 5–8 prompts your team actually used (copy-pasted), so the analysis is reproducible and gradable.

Reflection (write 100–150 words at the end of the brief)

Where did the AI tool genuinely surface something you wouldn't have caught by eyeballing the spreadsheet, and where did your team have to override or correct it? This is the part that's actually graded most closely — a plausible-sounding AI answer that the team didn't verify against the data will be marked down.